

Criterion	What We Assess	Excellent — Full Marks	Good — Most Marks	Fair — Partial	Poor / Missing	Marks
1. Data Preprocessing & Feature Engineering	Did the team properly prepare the data and engineer useful features?	Data cleaned and preprocessed systematically (missing values, outliers, encoding, scaling). Feature engineering is justified and improves the quality of model inputs. The preprocessing pipeline is reproducible.	Most preprocessing steps were applied correctly. Feature engineering is present, but some choices lack justification, or minor issues remain.	Basic preprocessing only (e.g., just dropping nulls). Little or no feature engineering; pipeline not clearly described.	Raw data used with no meaningful preprocessing. No awareness of data quality issues.	/6
2. Model Selection & Justification	Did the team choose appropriate models and justify their choices?	At least 3 models selected with clear justification tied to the problem type. Baseline model included. Hyperparameter choices explained.	Reasonable models chosen with partial justification. Baseline present or mostly clear reasoning given.	Models applied without adequate justification; no baseline; hyperparameters left at defaults with no acknowledgment.	Arbitrary or single model with no justification; no understanding of model suitability shown.	/6
3. Evaluation Strategy & Metrics	Are the right metrics used, and is the evaluation setup sound?	Metrics correctly chosen for the task (e.g., F1/AUC for imbalanced classification, RMSE/MAE for regression). Train/val/test split or cross-validation applied correctly. No data leakage. Metric choices explicitly justified.	Appropriate metrics used with minor issues (e.g., only accuracy on imbalanced data, or validation set not clearly separated).	Generic metrics used without justification. The evaluation setup has potential leakage or split issues.	Wrong metrics for the task, or no proper evaluation setup. Results are uninterpretable.	/6
4. Results & Comparative Analysis	Are results reported clearly and compared meaningfully across models?	Results reported for all models with correct metric values. Clear comparison table or chart. The best model is identified with reasoning. Results are reproducible (seed set, code available).	Results reported for most models. Comparison present but lacking depth or visualization.	Results reported for only one model, or the comparison is superficial (listing numbers without discussion).	Results missing, incorrect, or not comparable. No clear winner identified.	/6
5. Error Analysis & Critical Reflection	Does the team understand where and why their model fails?	Concrete error analysis: confusion matrix, misclassified examples, failure patterns. Team explains why errors happened.	Some error analysis present (e.g., confusion matrix shown) but limited discussion.	Errors acknowledged but no real analysis. Only high-level statements.	No error analysis. Team presents only positive results.	/4

6. Report Quality & Reproducibility	Is the report professional and well-structured, and is the work reproducible?	Figures/tables labeled and discussed. Code or methodology described well enough to reproduce results. Citations correct.	Mostly clear with minor formatting or citation issues. All figures captioned.	Disorganized sections or figures without captions. Reproducibility unclear.	Poorly written, missing sections, no figure labels, or no citations.	/2
TOTAL						/30

Key Expectations

- **Baseline first:** Always start with a simple baseline (e.g., majority class, linear model) before complex models.
- **At least 3 models:** Compare at least two different model families — not just the same model with different parameters.
- **No data leakage:** Preprocessing (normalization, feature extraction, etc.) must be fit only on training data, then applied to validation/test.
- **Metric choice matters:** Accuracy alone is not sufficient for imbalanced datasets. Justify every metric you report.
- **Error analysis is not optional:** Showing only good results is not scientific. We expect honest discussion of where and why models fail.
- **Reproducibility:** Set random seeds. Your results must be reproducible from your submitted code.

Team Contribution Table (must be included in your report)

Team Member Name	Student ID	Assigned Role	Main Responsibilities	Deliverables Contributed
Member 1	XXXXX	ML Lead / Coordinator	Model 1 selection, training runs, and result tracking	Model training & results section
Member 2	XXXXX	Data Engineer	Preprocessing pipeline, feature engineering	Data section & pipeline code
Member 3	XXXXX	Evaluation & Metrics Lead & Error Analyst	Evaluation strategy, metric selection, cross-validation	Evaluation section & Error analysis section
Member 4	XXXXX	ML Engineer	Model 2 selection, training runs, and result tracking	Model training & results section
Member 5	XXXXX	ML Engineer	Model 3 selection, training runs, and result tracking	Model training & results section

Each member is graded on the full report. Roles define primary responsibility, not exclusive ownership. All members are expected to contribute meaningfully across all sections and final presentation.